Fondamentaux de l'Optimisation – Travail de groupe

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1. Introduction

The objective of this work is to complete the introduction to the concepts covered during the course lectures regarding mathematical modeling and problem solving.

To achieve this, you will work in teams: This is not an individual task but a collaboration that should notably contribute to strengthening your organizational skills within a group.

The organization within the team is not the responsibility of the teaching staff, who may nevertheless intervene in case of difficulties or problems.

It is very important that each team member can highlight their role and contributions to the project's realization and completion.

These elements will be taken into account in the grading and may result in a different grade for each member.

The problem considered in this work is described in the next section.

2. Multi-Period Petrol Station Inventory-Routing and Loading- Scheduling Problems

Petroleum products are distributed from a central terminal to a set of petrol stations by a heterogeneous fleet of compartmented tank trucks. Each petrol station consumes several fuel products and has a separate storage capacity for each product. Replenishment decisions must therefore account simultaneously for inventory evolution at petrol stations, the allocation of fuel products to truck compartments, and vehicle routing.

In this project, students are invited to formulate, implement and computationally analyze **two related optimization problems** of increasing complexity.

The first problem focuses on the integration of inventory management, compartment allocation and vehicle routing over a multi-period planning horizon.

The second problem extends the first by explicitly considering the timing of operations within each day, customer time windows and limited loading resources at the depot.

The expected solution approach is **Mixed-Integer Linear Programming (MILP)** implemented using a mathematical programming solver such as CPLEX or Gurobi.

Students must themselves define all sets, parameters, decision variables, objective functions and constraints. The papers provided on Moodle may be used as references, but the proposed models must correspond exactly to the problems described below.

Problem 1 — Multi-Period Petrol Station Inventory-Routing Problem

A company operates a single fuel terminal supplying a set of petrol stations over a finite planning horizon divided into periods, for example days.

Several fuel products are considered. For each petrol station and each product, the demand or consumption during every period is known and deterministic. Each station has an initial inventory of every product, a minimum admissible inventory level representing the safety stock, and a maximum inventory level corresponding to the capacity of its underground storage tank.

Replenishment decisions are not fixed in advance. The model must determine **whether a station should be replenished in each period and how much of every product should be delivered**. A station does not necessarily have to be visited in every period. A delivery may be postponed if the available inventory is sufficient to cover future consumption.

This characteristic distinguishes the problem from a classical multi-period vehicle routing problem: inventory and replenishment timing are optimization decisions. This is also the central idea of the initial case-study definition.

Fuel is transported by a heterogeneous fleet of tank trucks. Each truck is divided into several compartments with known capacities. A compartment may contain only one fuel product during a trip and its content is assigned to at most one petrol station. A station may receive several compartments and may be supplied by several trucks or several trips; therefore, split deliveries are allowed. This corresponds to the multi-compartment structure already defined in the preliminary project.

Each active trip starts and ends at the depot. A truck may perform several trips during the same period, subject to a maximum number of trips.

The depot has sufficient supply of all fuel products.

Travel distances between the depot and petrol stations and between pairs of petrol stations are known.

The objective is to **minimize the total transportation distance over the complete planning horizon**, while guaranteeing that all station inventories remain within their admissible limits.

For this first problem, travel times, customer time windows, loading-bay limitations and explicit within-day scheduling are not considered.

Students must formulate this problem as a MILP model and implement it in a commercial optimization solver.

Problem 2 — Multi-Period Petrol Station Inventory-Routing and Loading-Bay Scheduling Problem

The second problem extends Problem 1 by explicitly modeling the operational schedule of each vehicle within every planning period.

All assumptions and decisions of Problem 1 remain valid. Consequently, the model must still simultaneously determine: the periods in which petrol stations are replenished, the quantities of every product delivered, the allocation of products and stations to truck compartments, the assignment of deliveries to trucks and trips, and the routes followed by the vehicles.

In addition, each planning period represents an operational working day during which all loading, travelling and delivery activities must be explicitly scheduled.

Each petrol station has an operating **time window** during which a delivery may take place. Travel times between all locations are known. A service duration is associated with deliveries at petrol stations. The schedule of a route must therefore guarantee that the truck can travel between consecutive locations and serve every visited station within its permitted time interval.

Tank trucks must be loaded at the depot before starting each trip. The depot contains a **limited number of loading bays**, shared by all vehicles. Each loading operation occupies one bay during its loading duration, and a loading bay cannot process two trucks simultaneously.

Loading duration may depend on the amount of fuel loaded. Appropriate loading rates and, if necessary, a fixed preparation time are provided in the instance data.

When a truck performs several trips during the same day, the trips must be chronologically compatible. In particular, a truck cannot start loading for a new trip before completing its preceding route and returning to the depot.

Every trip must be completed before the end of the corresponding working day.

Thus, in addition to the inventory, loading and routing decisions of Problem 1, Problem 2 must determine: **when every truck is loaded, which loading resource it uses, when it departs from the terminal, when each station is served, and when the truck returns to the terminal.**

The objective remains the minimization of total transportation distance. Scheduling and loading-bay constraints therefore restrict the set of operationally feasible replenishment plans rather than introducing a different objective.

Instances of Problems 1 and 2

The benchmark is publicly available: https://www.leandro-coelho.com/instances/multi-compartment-delivery/

Problem 1 must first be reduced to this special case by allowing only one trip per vehicle and period and by adapting any other assumptions necessary to obtain the corresponding benchmark variant.

Instances of Problem 2 can be generated from Androutsopoulos and Zografos (2024).

Expected work

1. **MILP** formulation of **Problems 1 and 2.**
2. Implementation of both formulations using CPLEX, Gurobi or an equivalent MILP solver.
3. Generation of instances Problems 1 and 2.
4. Analysis of solvers performance.

3. Evaluation

Several deliverables are expected in this work, some of which can be performed in parallel.

It is up to you to organize yourselves within the team, but it is (strongly) advised that everyone partially handles all aspects of the project (especially if you want the grading to be consistent).

The final grade for each person will be determined from several evaluations:

- A report,
- The source code,
- The results obtained,

3.1 Report

The report must be at the level expected of Master students. This means it must be well-structured, written, and presented.

Attention: no plagiarism will be tolerated. We have tools at our disposal via Moodle to detect such things, especially for reports. The goal of this part is to present, in your words, the methods you have chosen and how they work.

The section dedicated to presenting tests on the instances must also be prepared with the greatest care. You must show what your approach was and what conclusions you draw from it.

Finally, it is fundamental that your report clearly and precisely highlights the role and tasks performed by each member. A diagram illustrating the organization and progression according to the sessions can be a plus.

3.2 Source Code and Results Obtained

The evaluation of the source code is essentially based on the structure of the code, verification that it corresponds to the description in the report, and that it actually allows obtaining the reported results.

It is also expected that the code be well-documented in order to reproduce the tests.

3.3 Submission

All expected elements (report and source code) must be deposited on Moodle before the deadline, which will be strict and indicated on the course page.

It is strongly advised not to wait until the last minute to submit your work, to limit risks related to network problems, for example.

The submission will be in the form of an archive in .zip, .rar, or .7z format.

All modalities will be repeated on the Moodle course space, and compliance with them will be part of the grading.

REFERENCES

1. Coelho, L. C., & Laporte, G. (2015). Classification, models and exact algorithms for multi-compartment delivery problems. *European Journal of Operational Research*, 242(3), 854–864.
2. Cornillier, F., Boctor, F. F., Laporte, G., & Renaud, J. (2008). A heuristic for the multi-period petrol station replenishment problem. *European Journal of Operational Research*, 191(2), 295–305.
3. Cornillier, F., Laporte, G., Boctor, F. F., & Renaud, J. (2009). The petrol station replenishment problem with time windows. *Computers & Operations Research*, 36(3), 919–935.
4. Cornillier, F., Boctor, F., & Renaud, J. (2012). Heuristics for the multi-depot petrol station replenishment problem with time windows. *European Journal of Operational Research*, 220(2), 361–369.
5. Androutsopoulos, K. N., & Zografos, K. G. (2024). Modeling and solving the fuel distribution problem with unloading precedence and loading sequence considerations. *Annals of Operations Research*, 332, 909–947. This should be the